

Toheeb Eji

(647) 207-6299 | toheebeji@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

McMaster University | Hamilton, ON
B.S. in Computer Science - 3.78 GPA

Sep. 2025 - Present

TECHNICAL SKILLS

Languages: C#, Golang, JavaScript, Typescript, HTML/CSS, Python

Frameworks & Libraries: Node.js, React.js, Svelte, ASP.NET Core, TailwindCSS, FastAPI

Systems & Databases: PostgreSQL, SQLite

Developer Tools: Git, GitHub, Google Cloud Platform

PROJECTS

Sift | *Go, React, TypeScript, SQLite, BM25, Web Crawling*

Jul. 2026 - Aug. 2026

- Developed a high-performance search engine in Go, featuring a concurrent crawler with hostname-based rate limiting, dynamic worker pools, and graceful task cancellation.
- Built an inverted index utilizing BM25 ranking algorithms with relevance scoring weighted by title, domain, URL, and linguistic stemming for precise result coverage.
- Optimized search-time memory overhead by engineering a persistent binary index with memory-mapped posting files, facilitating efficient data access without full index loading.
- Achieved scalability for 500k pages, enhancing performance by 60% through frontier redesigns and DNS-failure caching, processing 20k URLs in under 2 minutes.

Serene | *React, Next.js, Typescript, TailwindCSS, C#, ASP.NET Core*

Jul. 2025 - Present

- Building a full-stack mental health platform using React/Next.js and ASP.NET Core with a PostgreSQL database.
- Engineered a semantic search recommendation engine using a vector database and OpenAI embeddings, implementing a custom RAG pipeline to surface relevant resources based on user demographics and struggle vectors.
- Integrated Google Gemini AI to generate context-aware prompts, utilizing Quartz.NET for reliable background job scheduling and content delivery.

Waypoint | *React, Mapbox GL JS, TailwindCSS, Tanstack Query, FastAPI*

Jan. 2026

- Architected FastAPI backend with async request handling, Monte Carlo simulation engine (1000+ agents), and integrated NASA SRTM elevation data, weather APIs, and OpenStreetMap features for terrain-aware movement
- Built real-time interactive map visualization using Mapbox GL JS, React, and TanStack Query to render predictive heatmaps and 12-hour forecasts with 500+ concurrent agent simulations.
- Deployed the server with Docker and Google Cloud Run for auto-scaling and production reliability.

HACKATHONS

DeltaHacks | *Hamilton, ON*

Jan. 2026

- Developed Waypoint - a search and rescue mission planning tool (see Projects)

RythmHacks | *Waterloo, ON*

Oct. 2025

- Winner of the Best Venture Track Award
- Developed StudyFlow - a mobile app for productive study breaks (see Projects)